

QIU-HAN GU

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Research Interests

Primary Interests: Formal methods for large language models, with a particular focus on abstract interpretation and its application in building trustworthy AI systems.

Other Interests: Neural Network Verification, Probabilistic Programming Verification, Constrained Generation Optimization.

Education

University of Illinois Urbana-Champaign

Master Science of Computer Science

Advisor: Prof. Gagandeep Singh

Aug.2024 – May.2026(Expected)

Champaign, United States

Nanjing University

Bachelor of Science in Computer Science and Technology

Advisor: Prof. Linzhang Wang and Prof. Yu Wang

Sept.2020 – Jun.2024

Nanjing, China

Preprints and Publications

Qiuhan Gu, Avaljot Singh, Gagandeep Singh, “Cost-Driven Synthesis of Sound Abstract Interpreters,” arXiv (under submission), 2025. [\[paper\]](#)

Qiuhan Gu, “LLM-Based Code Generation Method for Golang Compiler Testing,” Proceedings of the 31st ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering (ESEC/FSE), 2023. [\[paper\]](#)

Projects

Cost-Driven Synthesis of Sound Abstract Interpreters

Advisor: Prof. Gagandeep Singh

Jun.2025 – Nov.2025

FOCAL Lab, University of Illinois

- Proposed the first unified framework to automatically synthesize globally sound abstract interpreters across multiple abstract domains.
- Formalized the synthesis process as a constrained optimization problem over an infinite search space of abstract transformers and introduced a novel mathematically grounded cost function that quantitatively measures unsoundness deviation to guide refinement.
- Empirically demonstrated the framework’s capability to match handcrafted transformer quality and, more importantly, synthesize globally sound and precise transformers for complex nonlinear operators absent from the literature.

Enhancing DL System Fuzzing via LLMs

Advisor: Prof. Lingming Zhang and Prof. Darko Marinov

Sep.2024 – Dec.2024

University of Illinois

- Developed an automated pipeline to generate and validate deep learning operator constraints using large language models, improving operator validity from 46.3% to 57.6%.
- Designed a domain-specific language to represent operator constraints and incorporated symbolic verification techniques to ensure correctness.
- Integrated LLM-generated operators into NNSmith and NeuRI fuzzing frameworks, achieving higher coverage and discovering multiple compiler bugs.

LLM-Based Code Classification Method (Bachelor Thesis)

Advisor: Prof. Yu Wang

Feb.2024 – Jun.2024

SEG, Nanjing University

- Implemented an LLM-based high-quality code classification method of Go language, achieving an accuracy of 96.4%.

- Combine large language models and traditional methods to efficiently model code in terms of lexical, syntactic, and semantic aspects.
- Defined custom classification categories, providing fine datasets for downstream tasks.

LLM-Based Code Generation Method for Golang Compiler Testing

Sept.2023 – Dec.2023

Advisor: Prof. Yu Wang

SEG, Nanjing University

- Implemented a novel LLM-based high-quality code generation method for the Golang compiler, generating testcases with 3.38% average coverage and only 2.79% of them had syntax errors.
- Published a paper as the independent first author at ESEC/FSE Conference 2023, LLM-Based Code Generation Method for Golang Compiler Testing.
- Won 1st prize in the undergraduate division of ESEC/FSE'23 Student Research Competition.
- Kept on exploring the software testing technique and improving the performance of program analysis by machine learning.

Automatic Detection of Intracranial Aneurysms Based on Deep Learning

May.2022 – Apr.2023

Nanjing University

- Built a clinically applicable deep learning system for automated intracranial aneurysm detection from CTA images.
- Implemented and deployed a unified web platform allowing medical institutions to access and run the diagnostic system.
- Created a comprehensive implementation plan covering system design, deployment pipeline, and business integration.
- Won the first prize in the national competition (1/154)

Teaching Experiences

- University of Illinois Urbana-Champaign: CS427 Software Engineering, 2025 Fall
- University of Illinois Urbana-Champaign: CS101 Introduction to Computer Science, 2025 Spring
- University of Illinois Urbana-Champaign: CS427 Software Engineering, 2024 Fall
- Nanjing University: Advanced Python Programming, 2024 Spring
- Nanjing University: Advanced Python Programming, 2023 Fall

Honors & Awards

- 1st Prize, Undergraduate Division, ESEC/FSE'23 Student Research Competition, San Francisco, United States, Dec.2023
- Huawei Scholarship, Nanjing University, China, Oct.2023 (< 10%)
- 1st Prize, National College Student Entrepreneurship Competition, China, Mar.2023 (10%)
- National Scholarship of Undergraduate, Nanjing University, China, Dec.2021 (< 1%)